



Low-power, noninvasive measurement system for wearable ballistocardiography in sitting and standing positions



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ABSTRACT

Ballistocardiography (BCG), the noninvasive measurement of the reaction forces of the body to the cardiac ejection of blood, is one of few methods available for unobtrusively assessing the mechanical aspects of cardiovascular health outside clinical settings. In this study, **two sizes of piezoelectric film sensors were installed into the seat cushion of a chair and the foot insole, respectively, to acquire local BCG signals in sitting and standing positions.** The design and engineering implementation of the signal processing circuit and data transmission system are discussed below. Although wearable BCG measurement devices have been extensively researched to date, to the authors' knowledge, they have never been tested in regards to power consumption and electrostatic discharge (ESD) immunity. In this work, comprehensive experiments were conducted to show the low power consumption and high reliability of the proposed system. The shapes and the amplitudes of measured signals were examined in detail. The results suggest that the proposed system can feasibly and reliably be used to continuously record and evaluate cardiovascular changes occurring throughout an individual's daily life.

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1. Introduction

Modern ECG monitoring devices are wearable and have reliable accuracy, but they cannot function without disposable wet electrodes. The economical burden and skin irritation caused by such electrodes render wearable ECG equipment altogether undesirable as a long-term physiological monitoring product [1,2]. Ballistocardiography is a noninvasive method to record the ballistic force caused by the motion of heart and blood during each cardiac cycle. BCG changes differ between subjects, and thus may reflect the short- or long-term cardiovascular state of the individual wearer. Recent experimental studies involving subjects with various cardiovascular diseases and healthy subjects have shown that BCG signals can be used to trend cardiac output, changes in contractility, and beat-by-beat left ventricular function for arrhythmias [3]. BCG devices have been successfully integrated into everyday objects like beds [4], chairs [5,6], and weighing scales [3,7], allowing for unobtrusive long term monitoring of cardiovascular health.

Several researchers have demonstrated that certain BCG properties as measured via capacitive sensors (EMFi) differ significantly in different body postures. Sitting and supine positions are clearly distinguishable in the BCG signals as changed amplitudes and stable time intervals [8]. BCG-based systems in the horizontal position are advantageous in terms of sleep monitoring, which can facilitate sleep stage classification and may provide predictive data for the diagnosis of sleep-related disorders [9]. BCG signals measured from other parts of the body also have significant application value.

Physiological features can be detected by a variety of technologies to assess fatigue; electroencephalogram (EEG) data, for example, can be analyzed to determine variations in brain activity, while electrocardiogram (ECG) and photoplethysmography (PPG) data can be used to study heart rate variability (HRV) at progressive stages of fatigue [10,11]. Rogado et al. developed a fatigue detection system for automobile operators based on HRV from ECG during driving periods [12]. Gih Sung Chung et al. demonstrated an unconstrained BCG measurement system to be an appropriate substitute for the ECG to calculate HRV [13], where a force sensor underneath the driver's seat is the standard apparatus for measuring sitting BCG to monitor fatigue.

Measurements of the forces exerted upon the plantar surface of the foot are useful in sport and medical applications such as

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diagnostics and rehabilitation [14]. Polyvinylidene fluoride (PVDF) is a thin, elastic, and permanently charged semicrystalline piezoelectric film that can be installed in the smart insole to sense the force imparted by the physical exercise and systolic motion of plantar blood vessels. This allows the device to record the wearer's steps and heart rate (HR), as well as a robust estimation of his or her cardiac output, continuously during the exercise period [15].

In summary, BCG data acquired in sitting and standing positions merit further research. To date, however, there has been no study to specifically evaluate the differences between these two respective types of BCG signals. Moreover, low power consumption and electrostatic discharge (ESD) immunity tests seldom appear in the literature on wearable BCG equipments.

This paper presents a practical, reliable, and energy-efficient BCG measurement system consisting of three parts: 1) PVDF film sensors integrated into a seat cushion and foot insole; 2) data collection with analog front end (AFE); and 3) a data communication and interface component. BCG signals from six participants were measured in sitting and standing positions, and the experimental results were explored in detail regarding: 1) The power consumptions of the device; 2) the whole immunity against electrostatic discharge (ESD); and 3) the signal qualities produced by different situations. We conclude that the proposed system can feasibly and reliably acquire wearable BCG signals continuously for safety, health, and fitness monitoring.

2. Materials and methods

2.1. PVDF piezoelectric film sensors

PVDF polymer film is a pressure-sensing element with piezoelectric properties that create an electric charge on the welding electrodes upon sensing a change in pressure. Like most other piezoelectric sensors, the PVDF film cannot be used to measure static pressure due to slow discharging caused by electrical resistance. The minimum cut-off frequency of pressure changes is about 0.01 Hz and the suggested response time of the sensor is less than 0.1 ms [16–18].

As shown in Fig. 1, two types of PVDF film sensors were applied to the sitting and standing BCG measurement system:

- (1) One 0.028 mm thick PVDF film with double-sided metal coating guarded with a layer of antistatic fabric; the shielded cable has contact with the welding wire, which is conductive.
- (2) One 0.028 mm thick PVDF film embedded in two pieces of 0.125 mm thick polyester substrates riveted by two metal connectors.

As shown in Fig. 1, a PVDF film sensor 600 mm in length and 400 mm in width was placed under the thighs in the seat of an office chair to detect vibrations; a piece of PVDF film 25 mm in length and 13 mm in width was integrated into a normal foot insole.

Our cross experiments and repeated tests indicate that these two types of PVDF sensors have similar stability and response against mechanical deformations. Under normal temperature and humidity conditions, the proper shape and size of support at the rim of the foil result in a linear characteristic curve.

2.2. Circuit design and data collection

Fig. 2 shows the schematic circuits used to measure the small charge changes produced by the PVDF film. The output voltage (U_o) of the charge amplifier stage can be calculated as follows:

$$U_o = -\frac{1}{j\omega C_f} U_s = -\frac{C_s'}{C_f} U_s = -\frac{C_s' Q}{C_f C_s'} = -\frac{Q}{C_f} \quad (1)$$

where Q is the charge generated by pressure changes, C_s and R_s are the equivalent capacitance and resistance of the sensor, C_s' is the equivalent output capacitance, U_s is the equivalent input-signal voltage source, C_f is the feedback capacitance of the low input bias current (1 pA) preamplifier (OPA2336, Texas Instruments, Inc., Dallas, USA), and R_f is the feedback resistance added to prevent signal saturation.

A passive one-pole low-pass RC filter ($f_{-3dB} = 150$ Hz) was placed in series between the charge amplifier and the integrated analog front end (ADS1292, Texas Instruments, Inc., Dallas, USA). The ADS1292 is a dual-channel, simultaneously-sampling, 24-bit delta-sigma ($\Delta\Sigma$) analog-to-digital converter (ADC) with integrated programmable gain amplifiers (PGAs). During the BCG recording period, the ECG signal was simultaneously measured from the chest lead by AFE circuits at uniform sampling rate (250 Hz). The

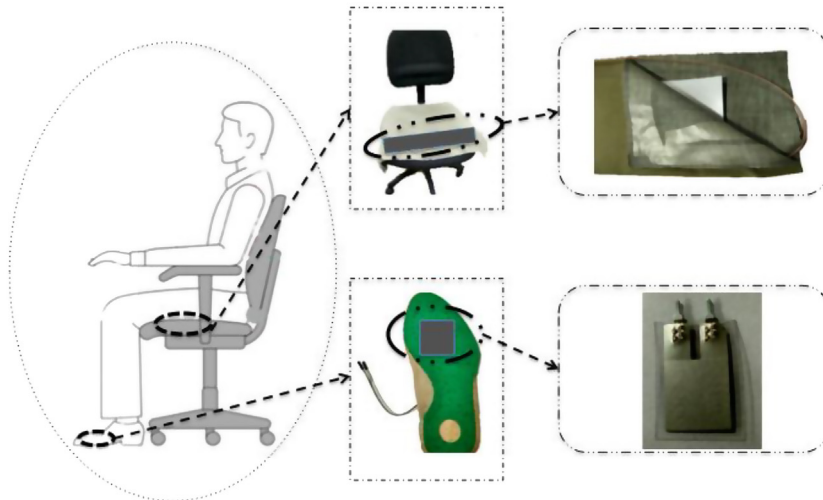


Fig. 1. Built-in PVDF film sensors and implementations of seat cushion and foot insole.

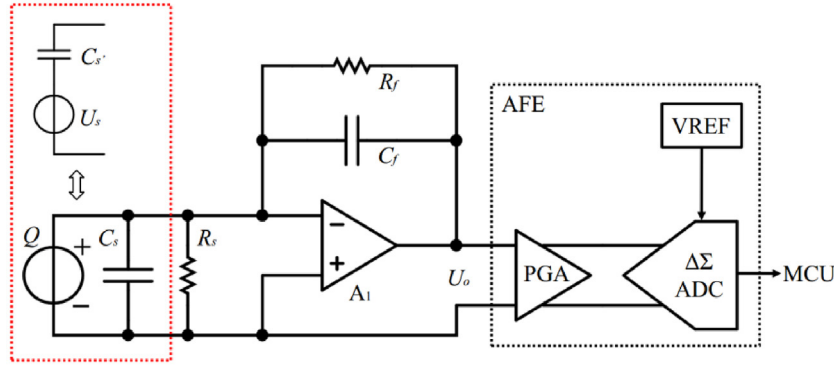


Fig. 2. Schematic circuits of BCG measurement based on PVDF film.

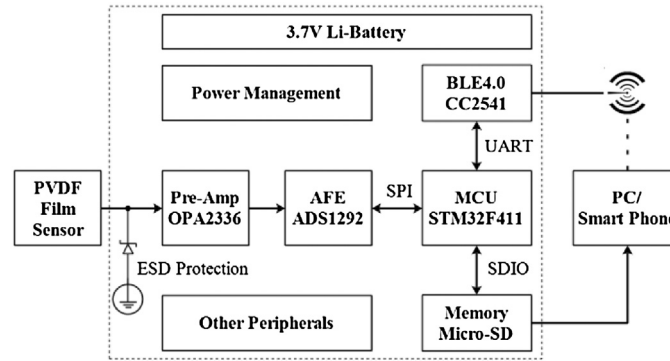


Fig. 3. Block diagram of proposed data collection and transmission system (DCTS).

BCG and ECG data packages from ADC were collected by the microcontroller (MCU) at the end of each sample with an SPI bus.

2.3. Data communication and interface

The complete data collection and transmission system (DCTS) is shown in block diagram form in Fig. 3. A Li-Battery (DC-3.7V, 180 mAh) provided clean power to the DCTS. The digitized output of the ADS1292 was connected to the input port of an STM32F411 (ST Microelectronics, Geneva, Switzerland), then the digitized signal was immediately stored into micro-SD card onboard with the secure digital input/output interface (SDIO). The real time data package can be wirelessly transmitted to a computer or smart phone using a Bluetooth Low Energy (BLE) module, which was configured in invisible UART mode with 57600 baud rate speed. Fig. 4 shows a photograph the DCTS hardware circuits which can measure BCG and ECG signals simultaneously. The final size of these hardware circuits could, in theory, be minimized further to suit more stringent requirements for wearable devices.

The BCG and ECG signals were digitally filtered by Butterworth band-pass filter in computer with the bandwidth of 0.1–Hz and 0.1–40 Hz, respectively.

2.4. Subjects and experiments

Six young participants with no history of cardiovascular disease participated in our experiment (Table 1). Their blood pressure (BP) and pulse were measured from the brachium of the left arm in seated positions just prior to each BCG measurement. Their BP and pulse were recorded before each plantar BCG measuring in standing positions as well at the beginning and ending of the

stepping test. The participants wore normal street clothes and socks during the experiment.

The experimental protocol was as follows for each case:

- (1) Subjects quietly rested on the experimental chair for 2 min and the sitting BP measurement was taken.
- (2) Subjects rested on the chair again for 5 min to acquire the sitting BCG signal, then, without removing the ECG electrodes, the same measurements were taken on the foot insole for 5 min in the standing position.
- (3) Subjects were instructed to perform a stepping movement test for 15 min to increase cardiac load and simulate walking. Once the test was finished, a second measurement for standing BP (standing BP 2) was taken.
- (4) Subjects stood on the foot insole again for the last 40 min of the test.



Fig. 4. Hardware circuits for BCG and ECG measurement.

Table 1

Gender, age, height (cm), weight (kg), blood pressure (mmHg) and heart rate (bpm) values in sitting and standing positions from subjects measured with a medical blood pressure monitor device. SBP is systolic blood pressure and DBP is diastolic blood pressure.

Subject No.	Demographics				Sitting BP			Standing BP 1			Standing BP 2		
	Gender	Age	Height	Weight	SBP	DBP	HR	SBP	DBP	HR	SBP	DBP	HR
1	M	27	183	80	118	61	67	117	58	70	145	83	87
2	M	24	175	63	123	66	73	121	65	71	134	66	99
3	M	29	165	63	109	68	86	112	62	87	124	77	100
4	F	24	172	62	120	78	72	111	73	78	132	67	71
5	F	21	165	56	104	59	65	116	56	71	123	66	98
6	F	24	162	48	114	66	81	106	56	82	128	72	92

Table 2

Power consumption of the three operation modes.

Mode	Analog Peripherals	AFE ADS1292	MCU STM32F411	Micro SD Card	BLE Module	Average Power Consumption	Lifetime
Standby	OFF	OFF	Standby	OFF	OFF	0.10 mW	>10 months
CR	ON	ON	Active/Stop	ON	OFF	1.71 mW	>15 days
RTC	ON	ON	Active/Stop	OFF	ON	26.3 mW	>24 h

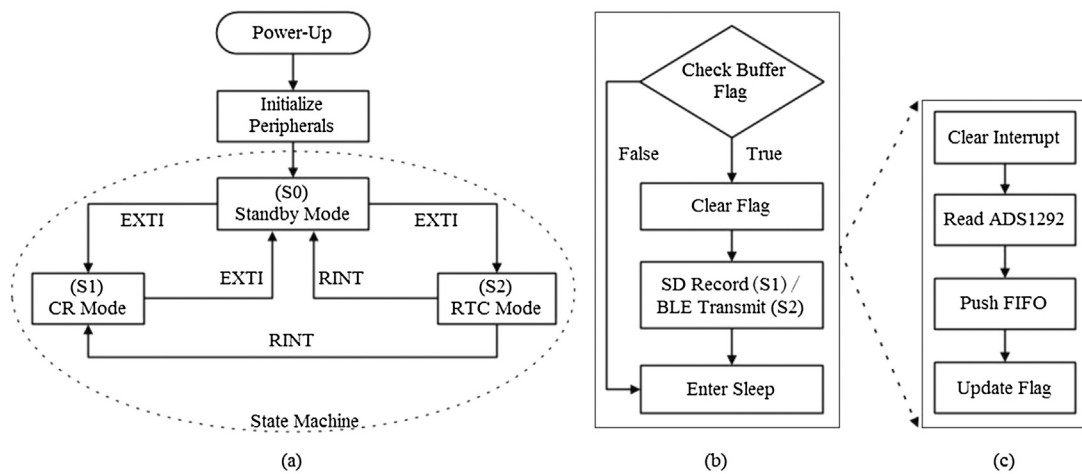


Fig. 5. Flow diagram of program execution. (a) Finite state machine. (b) Subroutine of CR/RTC mode. (c) Data-ready interrupt service routine of AFE.

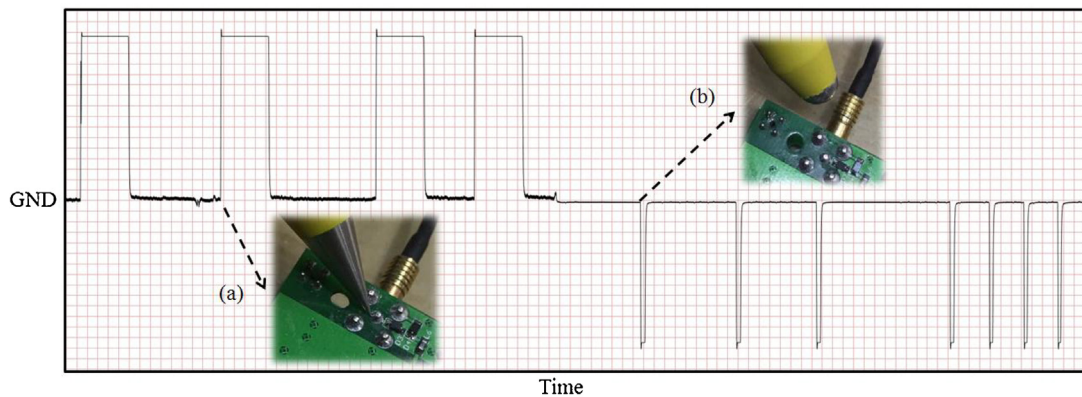


Fig. 6. Experiments of +8 kV electrostatic discharge. (a) On the input signal pin. (b) On the reference line.

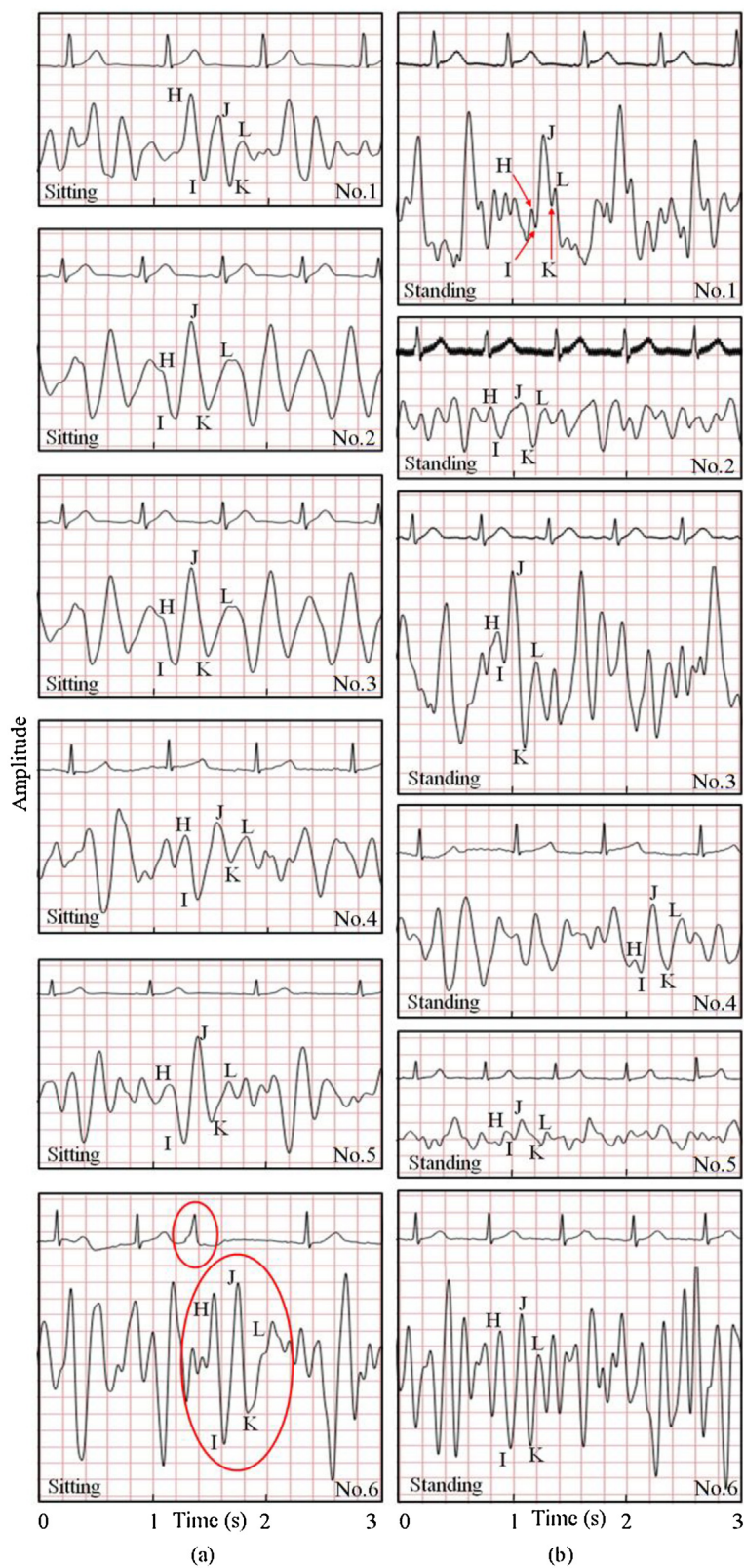


Fig. 7. Waveforms recorded from all subjects. (a) In seated positions. (b) In standing positions after stepping test.

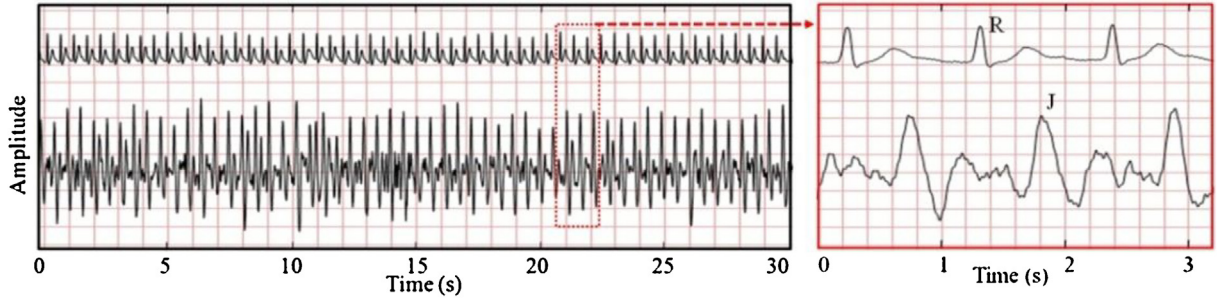


Fig. 8. Example waveforms recorded from participant No.1 after 10 min running exercise.

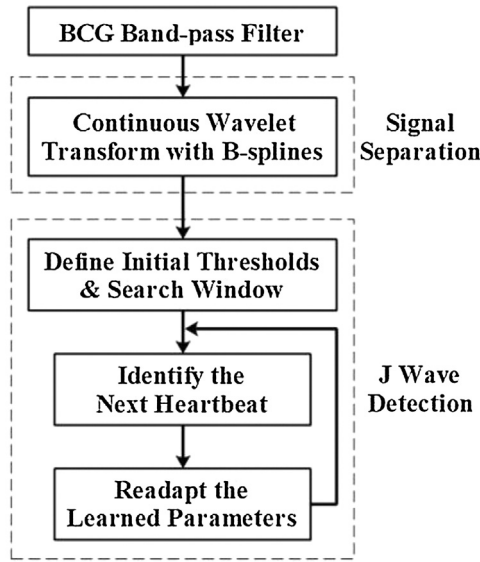


Fig. 9. Flowchart of the proposed heartbeat detect algorithm using BCG signals.

3. Results and discussion

3.1. Power consumption for DCTS

Including the standby mode, another two operation modes were provided to acquire signals for DCTS: Continuous recording (CR) mode and real-time communication (RTC) mode. After the system was powered up, the MCU configured specific registers of all the peripherals used and initialized them into low-power mode.

The average power consumption of DCTS then dropped to about 0.1 mW, and the system entered standby mode (S0), i.e., the original mode of the finite state machine (FSM). MCU was forced into CR mode or RTC mode using the different external interrupt (EXTI) commands. In these operation modes, MCU spent most of the time on sleep to conserve energy. As soon as the data-ready interrupt arose from AFE, MCU read the data package and pushed it into the FIFO area. After that, the program pointer returned back to the main process, and saved data into SD memory or transmitted it to the host device via the BLE module (if the buffer-full flag was checked true). The serial receive interruption (RINT) can be generated to switch operation states by BLE communication. All the steps are summarized in the flow diagram shown in Fig. 5.

Table 2 shows the functionalities of all modules and average power consumption depending on the working mode. A lithium battery (dc-3.7 V, 180 mAh) can supply the system with the energy of about 15 days and 24 h in CR and RTC modes, respectively.

3.2. Whole system ESD test

In the field of medical instruments, electrostatic discharge (ESD) immunity requirements for wearable electronic equipment are very strict. For instance, the international electro-technical commission (IEC) requires that ECG Holter devices shall comply with IEC60601-1-2 & IEC60601-2-47: A level of ± 6 kV shall apply for contact discharge to conductive accessible parts and coupling planes. Additionally, a level of ± 8 kV shall apply for non-conductive accessible parts and the equipment shall return to the previous operating mode within 10 s without loss of any stored data.

For wearable BCG measurement systems, static electricity caused by the friction between the body and the sensor interface is inevitable. When electric charge accumulates to a certain degree,

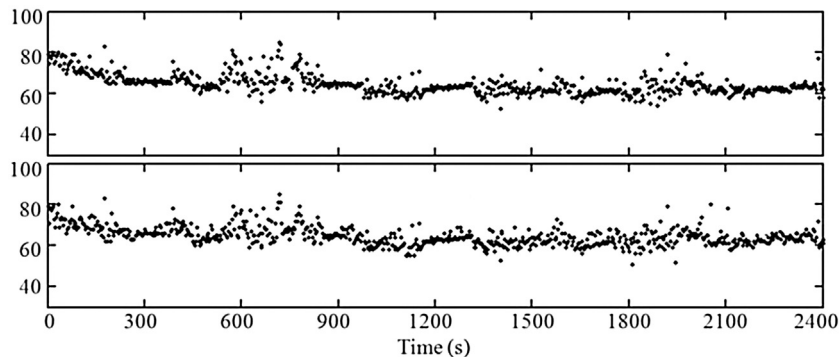


Fig. 10. BHRs (every 3 s) detected from BCG (bottom plot) are compared to an ECG reference (top plot) in participant No.5.

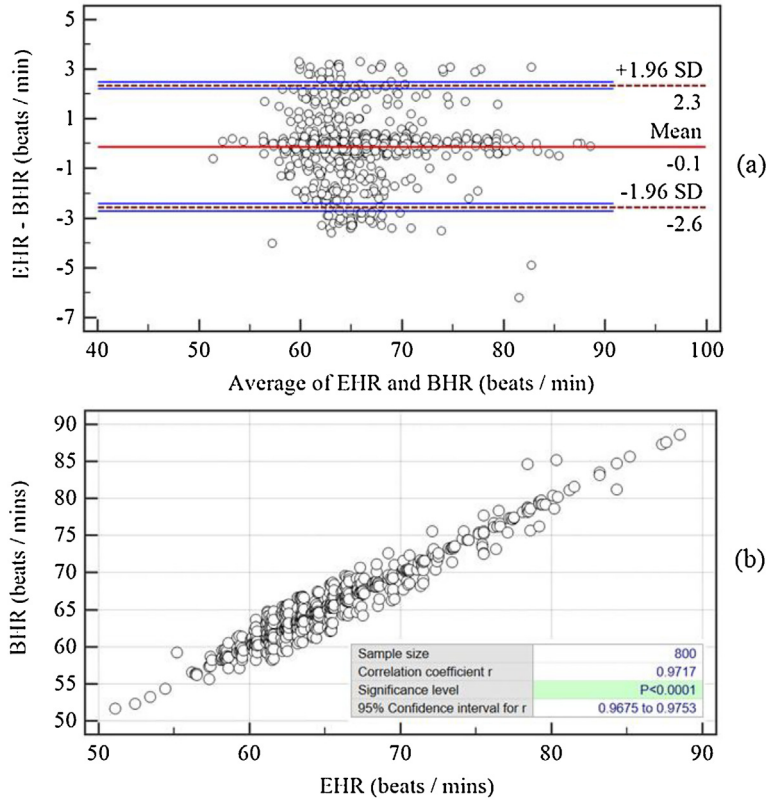


Fig. 11. Statistical analysis of average heart rates measured from BCG and ECG signals in participant No.5. (a) Bland-Altman plot between BHR and EHR. (b) Scatter diagram for correlation.

high-voltage power release phenomenon can be produced upon reaching the right discharge path. If the release is not inhibited, sensitive components may be compromised or damaged thus causing system failure or data loss.

A resistance and transient voltage suppressor (TVS) combinational network was utilized to limit peak current and peak voltage to protect the input terminals of the preamplifier. We used internal timer and independent watchdog embedded software to monitor the AFE module output. Once an exception occurred, data collection was interrupted immediately and the AFE unit or MCU was restarted to the steady state.

A simulator (NSG435, Teseq Group, Switzerland) was used to generate different levels of ESD pulses to shock the DCTS sensor interface. As shown in Fig. 6, after a group of +8 kV ESD pulses were applied for the input signal line and the reference line, four 2.5 s and seven 0.25 s square waves appeared in the measuring channel with the unloaded sensor. As lower impedance grew between the protection device and the protected line, faster energy absorption truncated the square wave duration.

Five prototypes were used for the discharge test, and each unit went through one hundred cycles of ± 8 kV discharge. Each prototype returned to the previous operating mode within 5 s without damage to any hardware unit or loss of any stored data, demonstrating that the proposed system meets the IEC standards on ESD immunity requirements.

3.3. Example waveforms from all subjects

When a person is in a seated position, the contact surface between the seat and human body is concentrated in the hip and

thigh. Placing a PVDF sensor between the hip and knee yields a BCG signal that is mainly characterized by the pressure pulsation of femoral and popliteal artery. The closer the sensor is to the knee joint, the smaller the attenuation effects of muscle and fat tissue on the signal. Fig. 7(a) shows the sample waveforms measured by six participants in seated positions. A clear BCG signal can be stably obtained by the DCTS system if the person is sitting still in the chair and a W-shaped complex consisting of the H-I-J-K-L peaks can be clearly recognized. Compared to the ECG, although the BCG signal did not distinguish between the locations of abnormal pacemakers, it clearly indicated the premature beats, cardiac arrest, and other arrhythmias.

The plantar arteries are mainly distributed in the bottom of the metatarsal and wrapped by ligaments and muscles. In normal conditions, the vascular pulsing signal captured by the sensor integrated in the insole is very weak. After a 15 min stepping training, the amplitudes of the BCG signal obtained by the insole were significantly enhanced as the heart rate and cardiac output of the participants increased. As shown in Fig. 7(b), although the signal sources were amplified by the additional exercise, the BCG signals in the standing position showed smaller amplitudes. With a more obscure H-I-J-K-L complex, the machine did not automatically identify plantar BCG signals as effectively. Further, notable differences in the load regulation ability of individual peripheral arteries prevented any enhancement in the amplitudes of BCG signals post-stepping test in participants No.2 and No.5.

The plantar artery runs between femoral and popliteal artery branches. Reduction in blood flow in the unit cross-sectional area and decrease in vessel wall elasticity significantly weaken the outgoing tension of the plantar artery compared to that of the leg femoral and popliteal artery. Amplitude attenuation and phase

delay of the mechanical wave can also be produced when the wave passes through the peripheral tissue and contacts the interface layer. Our experiments indicated that the measured signal amplitudes were almost consistent when a 1 cm thick medical silica gel pad was placed under the participant's feet or legs, however, which effectually indicates that the main cause of differences in amplitude between the two positions is the differences in artery contraction force.

The plantar BCG signals were measured by the DCTS system through the foot insole after participant No.1 ran for 10 min at a speed of 8 km/h, as shown in Fig. 8. The J-wave, which describes the acceleration of blood in the descending and abdominal aorta and the deceleration of blood in the ascending aorta [7] corresponds to the R-wave of the ECG; it is more easily detected due to the larger amplitude and distinctive peak pattern. When the DCTS system is further miniaturized and optimized, the insole device may be a very functional and wearable healthcare product yielding highly accurate number of steps and heart rate information. Future research is warranted to extract reliable characteristic parameters from wearable BCG signals and calculate the statistical correlation between physiological events and signals.

3.4. Heart rate detection under light motion artifacts

We found that the BCG signals acquired in standing positions were more susceptible to motion artifacts than those in seated positions, which created more noise. It is impractical to extract effective physiological information from the signals with interference via a computer in the conditions of intense exercise (e.g., running), but is possible to recognize the interference fragments and abandon them, or simply track the steps from those signals. In case of light exercise conditions, for example body swaying, the representative algorithm proposed by Alvarado-Serrano et al. [19] can be used to calculate the heart rate (HR) from the disturbed BCG signals. Flowchart of the signal processing steps is shown in Fig. 9.

In the signal separation stage, the fast implementation of the continuous wavelet transform with B-splines achieved a lower complexity than other algorithms [20–22]. In subsequent learning and decision stages, the J-wave detection was used to ensure low-complexity signal processing and heuristic rules.

For the last 40 min of the experiment, as described in Section 2.4, we adopted continuous processing for the BCG data as participants remained in standing positions, and calculated their average heart rate every 3 s. As shown in Fig. 10, the 800 heart rate values of each subject were defined as BHRs; 800 reference heart rate values defined as EHRs from the simultaneous recorded ECG were obtained using a CardioTrak Holter Analysis System (BeneWare, Hangzhou, China), a commercial medical software certified by CE. We applied a Bland-Altman plot [23] to assess the agreement between BHRs and EHRs for all subjects. The mean error of the average heart rates obtained from the worst result (subject No.5) was -0.1 beats/min and the 95% confidence interval (± 1.96 SD) was ± 2.6 beats/min, as shown in Fig. 11(a), which is still within the accuracy limits of ± 5 beats/min recommended by the Association for the Advancement of Medical Instrumentation (AAMI) standard for heart rate meters. The correlation value was 0.9714 ($p < 0.0001$). Fig. 11(b) shows a satisfactory scattered diagram.

4. Conclusion

Although current ECG devices are already sufficiently wearable and accurate, BCG allows cardiac activity to be monitored and recorded more simply and less obtrusively. Using two sizes of PVDF film sensors, we designed an energy-friendly and jam-proof

system to measure local BCG signals in sitting and standing positions. The equipment can record the complete BCG signals for 15 days continuously and resist a level of ± 8 kV ESD pulse repeatedly without any data loss. Compared to BCG signal under the thigh, the plantar BCG signal has smaller amplitude and is more susceptible to motion artifact and postural sway. Based on results obtained from six study participants, we demonstrated that the proposed algorithm as-applied to the noninvasive system meets the accuracy limits recommended by AAMI for beat-to-beat heart rate detection.

By implementing the measurement system that is already commercially available for the seat cushion and foot insole, we believe that the proposed solution could be streamlined for health, fitness, and rehabilitation medicine applications.

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